

12 — Sail On: Wind 10 m/s @ 90° Shrinks the Cascade

Proves · Sail: an intervention applied BEFORE the cascade, which shrinks the swing itself.

Mechanics this scenario turns on

- Sail thrust is subtracted from propulsion **before** the cascade runs, so it shrinks the swing itself — not just the average load.
- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\Sigma J + \Sigma L = \Sigma H$ — the sea sets the swing, the battery moves the split.
- Battery Benefit runs the pipeline **twice**: world A (demand = average + uncovered L) and world B (demand = average + the full swing H). $\text{Benefit} = (\text{FOC}_B - \text{FOC}_A) \times \text{hours}$. Both are **optima**; a pinned baseline is ignored. Unticking "Enable Battery" in the UI gives a *third* world — raw demand, swing carried by nobody — which burns less than A and is not the comparison.

Panels described below · The wind → cascade chain · Power Demands · Baseline & IL1 · Battery Benefit — and why it FELL

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 01.

Test 01's vessel + sail enabled, true wind 10 m/s at 90°, vessel 12.5 kn.

The wind → cascade chain (test point B8)

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Wind (speed+angle) + vessel speed → apparent wind → sail thrust = 539.93 kW
Sail-adjusted propulsion avg = 11 463 - 540 = 10 923 kW
Cascade uses the ADJUSTED value:  $H = 10\,923 \times 5\% = 546.2$  (was 573.15!)
→  $I = 546.2$  →  $J = 546.2 \times 0.35 = 191.2$  →  $L = 355.0$ 
Hotel unchanged: 76 → 3.8 / 72.2 (wind doesn't shake the galley)
Tiles: PS = 191.2+3.8 = 195 · SR = 355+72.2 = 427.2
```

Power Demands (new column: SAIL REDUCTION)

ME would carry $11\,463 + 355 = \mathbf{11\,818}$; sail takes **-540** → net **11 278**. ME total = $11\,278 + 3\,250 = \mathbf{14\,528}$ (header) · Energy $14\,528.1 \times 5\,000 = \mathbf{72\,640\,366}$. IL1 Optimization Details shows: "Sail Contribution: $11\,463 \rightarrow 10\,923$ kW · Sail Power 540 kW (4.7 %)".

Baseline & IL1

All combo FOCs drop (lighter ME): 2.5631 ... 2.6072; baseline (3rd-from-worst) 2.5673 → **12 836.6 t/yr**; IL1 2.5631 → **12 815.4**; savings 21.2.

Battery Benefit — and why it FELL

Canonical backend values at \$780: **165.5 t/yr = \$129 105**. The sail and the battery compete for the same swing: the covered band shrank (195 vs 204.4), so the battery's relative value dropped from 173.7 — even though the ship as a whole burns much less (baseline 12 837 vs 13 307). Mirror image of test 16, where heavier sea inflates both.

⚠ On the live screen this test displayed **\$132 416** = benefit × **800** while IL1 costs used 780 — two responses mixed on one screen. Cause: the profile-restore double-fire (finding #5) let an early calculation run with the form DEFAULT fuel price (constant `DEFAULT_VALUES.FUEL_PRICE = 800` , which also explains the "\$800/ton" Financial badge — finding #4). The backend math itself is consistent; the display race is the bug.

Takeaway: sail power feeds the cascade (sail-adjusted propulsion), shrinking both what the battery covers and what the gensets reserve.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-07 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)